



ZETHETA ALGORITHMS PVT. LTD.

Data Analyst – Securitisation AI Agent Internship Programme

AUTO LOAN ABS SECURITISATION RISK ANALYTICS USING POWER BI

A Project Report Submitted for the
Data Analyst – Securitisation AI Agent Industry Project

Submitted by

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July 2026



DECLARATION

I, Payal Dilip Nikam, hereby declare that this project report titled "Auto Loan ABS Securitisation Risk Analytics using Power BI" is my own original work, carried out as part of the Data Analyst – Securitisation AI Agent Industry Project with ZeTheta Algorithms Pvt. Ltd..

I further declare that the dashboards, DAX measures, Row-Level Security configuration, and Excel cross-validation presented in this report were developed and tested by me, and that all data used is either synthetic or anonymised sample data provided solely for the purpose of this learning engagement. This report has not been submitted, in whole or in part, for any other project, award, or publication.

Date: July 2026

Payal Dilip Nikam



ACKNOWLEDGEMENT

I would like to express my sincere gratitude to ZeTheta Algorithms Pvt. Ltd. for providing me the opportunity to work on the Data Analyst – Securitisation AI Agent internship project. This engagement gave me valuable hands-on exposure to auto loan asset-backed securitisation, credit risk analytics, IFRS 9 expected credit loss modelling, and enterprise Power BI dashboard development.

I am thankful to the project mentors and reviewers at ZeTheta Algorithms Pvt. Ltd. for their structured project methodology, the DAX training material, and the detailed specification that guided every stage of this work — from dataset understanding and data modelling to dashboard design, DAX measure creation, Row-Level Security, and Excel-based cross-validation.

Finally, I acknowledge the broader learning resources — including Microsoft's official Power BI and DAX documentation — that supported my understanding of the concepts applied throughout this project.

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ABSTRACT

Project Overview

Asset-backed securitisation converts a pool of illiquid auto loan receivables into tradeable, tranchised securities, and its ongoing surveillance requires continuous, multi-dimensional monitoring of credit risk, delinquency, cash-flow waterfalls, and regulatory expected-credit-loss provisioning. This project builds an end-to-end Power BI analytics solution that replicates the surveillance workflow used by originators, servicers, rating agencies, and investors in an auto loan ABS transaction, using a simulated 500-loan pool as the underlying dataset.

Objectives

- To design a star-schema data model connecting Loan Master, DPD History, Monthly Loss, Vintage Data, Tranche Data, Stress Testing Data, and Calendar tables.
- To build approximately 40 DAX measures covering pool balance, weighted portfolio metrics (WAC, WAM, WALA), credit risk, IFRS 9 ECL, waterfall, and stress-testing analytics.
- To develop ten interconnected Power BI dashboards covering the full securitisation lifecycle, from an executive overview to rating-agency data tape export.
- To implement Row-Level Security for five stakeholder roles and independently validate key measures against manual Excel calculations.

Technology Used

Microsoft Power BI Desktop (data modelling, DAX, visualisation), Power Query (ETL and data preparation), Microsoft Excel (independent cross-validation of measures), and DAX as the primary calculation language.

Final Outcome

The project delivers a fully functional ten-page Power BI report, a validated DAX measure library, a configured Row-Level Security framework across five roles, and an Excel-based validation trail confirming the accuracy of the core KPIs — together forming a production-style securitisation surveillance solution suitable for portfolio managers, risk analysts, auditors, servicers, and senior management.



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ABBREVIATIONS

Abbreviation	Full Form
ABS	Asset-Backed Security
DPD	Days Past Due
IFRS 9	International Financial Reporting Standard 9
LGD	Loss Given Default
PD	Probability of Default
EAD	Exposure at Default
ECL	Expected Credit Loss
WAC	Weighted Average Coupon
WAM	Weighted Average Maturity
WALA	Weighted Average Loan Age
LTV	Loan-to-Value Ratio
DTI	Debt-to-Income Ratio
CIBIL	Credit Information Bureau (India) Limited Score
RLS	Row-Level Security
CSV	Comma-Separated Values
DAX	Data Analysis Expressions
ETL	Extract, Transform, Load
SPV	Special Purpose Vehicle
NPA	Non-Performing Asset
SMA	Special Mention Account
RBI	Reserve Bank of India
GFC	Global Financial Crisis
IL&FS	Infrastructure Leasing & Financial Services Ltd.



CHAPTER 1

Introduction

1.1 Securitisation

Securitisation is the financial process by which a pool of illiquid, income-generating assets — such as auto loans, mortgages, or credit card receivables — is aggregated and transferred to a Special Purpose Vehicle (SPV), which then issues tranchised securities backed by the cash flows of that pool to capital-market investors. The originating lender obtains liquidity and transfers credit risk off its balance sheet, while investors gain access to a diversified stream of principal and interest cash flows, prioritised across a Senior, Mezzanine, and Equity tranche structure. Credit enhancement mechanisms — overcollateralisation, subordination, and cash reserves — protect senior tranches from pool-level losses.

1.2 Auto Loan ABS

An Auto Loan Asset-Backed Security (Auto Loan ABS) is a securitisation structure backed specifically by a pool of retail or commercial vehicle loan receivables. Auto loan pools are typically granular — hundreds to thousands of individual loans — with well-understood amortisation and recovery behaviour, which makes them a preferred asset class for both originators seeking balance-sheet relief and investors seeking predictable, short-duration cash flows. Surveillance of an Auto Loan ABS transaction requires monthly tracking of pool balance run-off, delinquency (DPD) migration, vintage cohort performance, and waterfall allocation across tranches, all of which this project's analytics solution is designed to support.

1.3 Business Problem

Once an auto loan pool is securitised, originators, servicers, trustees, rating agencies, and investors each require a different, role-specific view of the same underlying pool — yet the raw servicing data (loan-level balances, DPD history, monthly collections, tranche cash flows, and stress-scenario outputs) arrives as disconnected flat files. Manually reconciling these files each month across multiple stakeholder groups is slow, error-prone, and does not scale as the number of pools grows. There is a need for a single, governed, self-service analytics layer that can compute standardised risk and performance measures once, and then present them differently to each



stakeholder — Portfolio Manager, Risk Analyst, Senior Management, Auditor, and Servicer — while remaining independently verifiable against manual calculation.

1.4 Objectives

- Consolidate seven source tables (Loan Master, DPD History, Monthly Loss, Vintage Data, Tranche Data, Stress Testing Data, Calendar) into a single star-schema Power BI data model.
- Design and validate a library of ≈ 40 DAX measures spanning pool balance, weighted portfolio statistics, credit risk, IFRS 9 ECL, waterfall, and stress-testing analytics.
- Build ten role-relevant, interconnected dashboards covering the complete securitisation surveillance lifecycle.
- Implement Row-Level Security across five stakeholder roles to demonstrate governed, role-based access control.
- Independently cross-validate key DAX measures against manual Excel calculations to confirm computational accuracy.

1.5 Scope

The scope of this project is limited to the analytics and reporting layer built on top of a simulated 500-loan auto loan pool, using synthetic/sample data representative of an Indian auto loan ABS transaction. The project follows a 15-day, phase-wise project methodology — spanning dataset understanding, data preparation, data modelling, DAX measure development, dashboard build-out, Row-Level Security configuration, and Excel-based validation — as prescribed under the Data Analyst – Securitisation AI Agent project framework. Live integration with a production loan servicing system, real investor cash settlement, and legal/trust deed administration are outside the scope of this project.

1.6 Expected Outcome

On completion, the project delivers a validated, ten-dashboard Power BI report supported by a documented DAX measure library, a working Row-Level Security model, and an Excel validation trail — providing a realistic demonstration of how a securitisation surveillance analytics solution is designed, built, and quality-checked in an industry setting.



1.7 Alignment with the Project Specification (Parts A–F)

This project was executed against ZeTheta's 96-page “Data Analyst – Securitisation AI Agent” specification document, which is organised into six parts (Part A through Part F). The table below maps each part of the specification to where it is addressed in this report.

Part	Specification Content	Addressed In
Part A	DAX training material — core DAX functions, filter context, and calculation patterns used to build measures.	Chapter 4 (Data Preparation) and Chapter 6 (DAX Measures)
Part B	Gamified simulation concept — scenario-based learning approach used to understand stress-testing and waterfall mechanics before implementation.	Chapter 2.6 and Chapter 7.8 (Stress Testing & Investor Dashboard)
Part C	Case studies — 2008 Global Financial Crisis and the 2018 IL&FS default, illustrating securitisation and credit-risk failure modes.	Chapter 2.7 (Case Studies)
Part D	15-day project methodology — phase-wise execution plan from dataset understanding to final validation.	Chapter 1.5 (Scope) and Chapter 4 (Data Preparation)
Part E	No-code tools guide — Power Query and Power BI Desktop as no-code/low-code ETL and visualisation tools.	Chapter 4 (Data Preparation) and Chapter 7 (Dashboard Development)
Part F	Submission protocol — report structure, screenshot evidence, and validation requirements for final submission.	Chapters 8–9 (Excel Validation, RLS) and this report's overall structure



CHAPTER 2

Business Understanding

2.1 Loan Portfolio

The underlying asset pool consists of 500 individual auto loans (LoanID series ZAAUTO2024-1-00001 through ZAAUTO2024-1-00500), spanning vehicle types — SUV, MUV, Sedan, Compact SUV, and Hatchback — and originated across states including Madhya Pradesh, Maharashtra, Andhra Pradesh, Kerala, Karnataka, West Bengal, Chhattisgarh, Delhi, Telangana, Odisha, Punjab, and others. Each loan record carries borrower-level attributes (CIBIL score, DTI ratio), loan-level attributes (current balance, LTV, interest rate, months on book), and risk attributes (PD, LGD, EAD) that together drive every downstream measure in the report.

2.2 Credit Risk

Credit risk in the pool is monitored through DPD (Days Past Due) bucketing — Current, 1–29, 30–59, 60–89, 90–119, and 120+ DPD — mapped to the RBI's SMA (Special Mention Account) classification (SMA-0, SMA-1, SMA-2, and NPA). The Credit Risk & DPD Analysis dashboard tracks overdue amount, cure rate, overdue accounts, and repossession count, alongside state-wise and vehicle-type-wise overdue concentration, giving the Risk Analyst an early-warning view of pool deterioration.

2.3 Investor Reporting

Investors holding Senior, Mezzanine, or Equity tranche notes require monthly visibility into principal and interest cash flow (paid versus due), current tranche balance allocation, credit enhancement levels, coupon rates, and trigger status (Pass / Watch / Triggered). The Waterfall & Tranche Analysis dashboard and the Stress Testing & Investor dashboard together serve this reporting need, presenting investor yield, PD multiplier, recovery rate, and scenario-wise stress loss.

2.4 Rating Agency Reporting

Rating agencies require a standardised, loan-level “data tape” — borrower ID, vehicle type, state, current balance, current LTV, and CIBIL score — together with pool-level summary statistics, to support ongoing surveillance ratings and periodic rating actions. The Rating Agency Reporting

dashboard and Data Tape Export module in this project reproduce this standard rating-agency output format, including pool performance by state, loss and recovery trends, delinquency distribution, and prepayment trend.

2.5 IFRS 9

Under IFRS 9, loans are classified into three stages based on credit deterioration since origination — Stage 1 (performing), Stage 2 (significant increase in credit risk), and Stage 3 (credit-impaired) — with Expected Credit Loss (ECL) computed as a function of Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). The IFRS 9 ECL Analytics dashboard in this project computes Total ECL, Average PD, Average LGD, Average EAD, stage migration trends, and ECL provision by state and by servicer.

2.6 Stress Testing

Stress testing evaluates how the pool and its tranches would perform under adverse macroeconomic scenarios — Base, Moderate, Severe, and Crisis — by applying scenario-specific PD multipliers and recovery-rate haircuts to the pool. The Stress Testing & Investor dashboard in this project reports stress loss across all four scenarios, expected loss versus stress loss by pool, collection efficiency, portfolio rating distribution, and pool-level trigger status (Pass / Triggered / Watch) under each scenario — mirroring the scenario-simulation approach introduced in Part B of the project specification.

2.7 Case Studies: Lessons from Structured Finance Crises

2.7.1 The 2008 Global Financial Crisis

The 2008 Global Financial Crisis originated substantially in the US mortgage-backed securitisation market, where subprime mortgage pools with deteriorating underwriting standards were repackaged into highly rated mortgage-backed securities and collateralised debt obligations. As underlying borrower defaults rose, the assumed diversification benefit of pooling failed to materialise because loan performance was correlated by a common macroeconomic shock (a nationwide housing price decline), rather than being independent across loans. The crisis exposed the danger of relying on static, point-in-time credit ratings without continuous, granular surveillance of pool-level delinquency and vintage performance — precisely the kind of ongoing monitoring this project's Vintage Analysis and Credit Risk dashboards are designed to provide.



2.7.2 The IL&FS Default (2018)

In India, the 2018 default of Infrastructure Leasing & Financial Services Ltd. (IL&FS) — a systemically important non-banking financial company involved in infrastructure financing and structured debt — triggered a severe liquidity crunch across the broader NBFC and securitisation-linked funding market. The episode highlighted the importance of granular, servicer-level and trigger-based monitoring (rather than reliance on aggregate ratings alone), reinforcing the rationale for the trigger-status tracking (Pass / Watch / Triggered) and servicer-wise ECL breakdown implemented in this project's Waterfall, Stress Testing, and IFRS 9 dashboards.

Both case studies reinforce the same operating principle that underpins this project: securitisation surveillance must be continuous, loan-level, and scenario-aware, rather than a one-time, pool-level assessment at issuance.

CHAPTER 3

Dataset Description

The data model is built from seven source tables. Each table is described below with its key columns, business purpose, and its relationship to the central Loan Master (fact) table.

3.1 Loan Master

Grain: one row per loan (500 rows). Acts as the primary fact/dimension hub of the model.

Column	Purpose
LoanID, BorrowerID	Unique loan and borrower identifiers
VehicleType, State, LoanPurpose	Loan segmentation attributes
OriginalLoanAmount, CurrentBalance	Pool balance and outstanding exposure
InterestRate, MonthsOnBook, RemainingTerm	Coupon and seasoning attributes (drive WAC, WALA, WAM)
LTV_Current, DTI_Ratio, CIBIL_Score_Current	Credit quality attributes
PD_Estimate, LGD_Estimate, EAD, ECL_Provision	IFRS 9 risk parameters
TotalPaymentsMade, TotalPaymentsDue	Drives Collection Efficiency

Relationship: one-to-many to DPD History (by LoanID); connects to Calendar via a reporting-date relationship for time intelligence.

3.2 DPD History

Grain: one row per loan per reporting period, tracking delinquency status over time.

Column	Purpose
LoanID, ReportingDate	Links to Loan Master and Calendar
CurrentBalance, AmountOverdue, EMIsOverdue	Delinquency exposure measures
DPD_Bucket, RBI_SMA_Class	Delinquency and regulatory classification



Column	Purpose
CureFlag, RepossessionFlag, WriteOffFlag	Loan-level workout status flags

Relationship: many-to-one to Loan Master (by LoanID); many-to-one to Calendar (by ReportingDate).

3.3 Monthly Loss

Grain: one row per pool per calendar month, summarising portfolio-level loss and collection performance.

Column	Purpose
ReportingDate, PoolID	Links to Calendar and pool dimension
GrossLoss_ThisMonth, NetLoss_ThisMonth	Monthly gross and net charge-offs
Recoveries_ThisMonth	Post-default recovery cash flow
CollectionEfficiency, EOP_Balance	Monthly collection performance and end-of-period balance

Relationship: many-to-one to Calendar (by ReportingDate).

3.4 Vintage Data

Grain: one row per origination vintage cohort per month-on-book, enabling cohort (static-pool) analysis.

Column	Purpose
VintageID, MonthOnBook	Cohort identifier and seasoning index
OriginalPoolBalance, RemainingPoolBalance	Pool factor / amortisation tracking
CumulativeNetLossRate, MarginalLossRate	Cumulative and incremental loss curves
CurrentDelinq30Plus	30+ DPD rate by cohort



Relationship: standalone analytical table, filtered independently by VintageID and Month on Book.

3.5 Tranche Data

Grain: one row per tranche (Senior, Mezzanine, Equity) per reporting date.

Column	Purpose
TrancheName, ReportingDate	Tranche identifier and period
CurrentBalance, CouponRate	Tranche outstanding balance and coupon
CreditEnhancement	Subordination / credit-support percentage
PrincipalPaid, PrincipalDue, InterestPaid, InterestDue	Waterfall cash flow tracking
TriggerStatus	Pass / Watch / Triggered performance trigger

Relationship: many-to-one to Calendar (by ReportingDate); linked conceptually to the pool via PoolID.

3.6 Stress Testing Data

Grain: one row per pool per stress scenario (Base, Moderate, Severe, Crisis).

Column	Purpose
PoolID, Scenario	Pool and scenario identifiers
PD_Multiplier, RecoveryRate	Scenario stress parameters applied to base PD/recovery
PortfolioBalance, Rating	Scenario-level portfolio balance and credit rating
InvestorYield, TriggerStatus	Scenario investor return and Pass/Watch/Triggered flag

Relationship: many-to-one to a Pool dimension; scenario is used as a slicer across the Stress Testing dashboard.

3.7 Calendar

Grain: one row per calendar date, spanning the full reporting horizon of the pool.



Column	Purpose
Date, Year, Month, MonthName, Quarter	Standard time-intelligence attributes
ReportingDate flag	Marks month-end reporting dates used for snapshot measures

Relationship: one-to-many to DPD History, Monthly Loss, and Tranche Data (by date); enables consistent time-intelligence (MTD, YTD, trend) calculations across every dashboard.



CHAPTER 4

Data Preparation

Data preparation was carried out using Power Query — the no-code/low-code ETL tool referenced in Part E of the project specification — within Power BI Desktop, ensuring every transformation step is recorded, auditable, and repeatable without writing procedural code.

4.1 Power Query

Each of the seven source tables (Loan Master, DPD History, Monthly Loss, Vintage Data, Tranche Data, Stress Testing Data, and Calendar) was imported through Power Query as a separate query, with a dedicated applied-steps sequence per table so that transformations remain independently traceable and re-runnable when the underlying source data refreshes.

4.2 Data Cleaning

4.2.1 Duplicate Removal

LoanID (Loan Master) and the composite LoanID + ReportingDate key (DPD History) were checked using the Remove Duplicates step to ensure a strictly one-row-per-loan and one-row-per-loan-per-period grain, which is required for the DISTINCTCOUNT and SUMX-based DAX measures used throughout the report to return correct results.

4.2.2 Missing Values

Numeric fields such as RecoveryAmount, EMIsOverdue, and CreditEnhancement were checked for blanks and replaced with 0 where a blank genuinely represents “not applicable” (e.g., no recovery has yet occurred), while blanks in categorical fields such as DPD_Bucket and RBI_SMA_Class were reviewed and, where necessary, defaulted to “Current” / “Standard” to avoid inflating risk-bucket counts.

4.2.3 Data Type Conversion

Reporting dates were explicitly converted to the Date type (rather than left as text) to enable the Calendar table's mark-as-date-table relationships and time-intelligence DAX functions; monetary columns were converted to Decimal Number/Fixed Decimal, and flag columns (CureFlag, RepossessionFlag, WriteOffFlag) were converted to the True/False (Boolean) type so they could be used directly inside CALCULATE filter arguments.



4.3 Data Transformation

- Unpivoted/reshaped monthly waterfall columns in Tranche Data so that Principal Paid, Principal Due, Interest Paid, and Interest Due could be plotted as parallel monthly series.
- Derived the LTV Band and LTV Band Sort columns (via conditional/SWITCH logic, later implemented as DAX calculated columns) to support the Pool Stratification Analysis dashboard's banded charts.
- Standardised State and VehicleType text values (trimmed whitespace, consistent casing) so that slicers and the state-wise map visuals group correctly rather than splitting a single state into multiple slicer entries.
- Converted all balance-type columns from absolute Rupees to Crores at the reporting layer via DAX DIVIDE(..., 10000000) measures, rather than in Power Query, so that the underlying detail-level values remain available for row-level validation.



5.2 Table Roles in the Model

Table	Role in Model
Loan Master	Primary dimension table — one row per loan; carries loan, borrower, and IFRS 9 risk attributes. Every loan-level measure (Total Pool Balance, WAC, WAM, WALA, Weighted Avg LTV, Total ECL) is computed directly from this table.
Calendar	Shared time-intelligence dimension — marked as the official Date table so that all trend, MTD, and YTD calculations behave consistently across DPD History, Monthly Loss, and Tranche Data.
DPD History	Fact table for delinquency — related many-to-one to Loan Master; drives Credit Risk & DPD Analysis dashboard measures (Cure Rate, Overdue Accounts, Repossession Count).
Vintage Data	Standalone cohort-analysis fact table; drives the Vintage Analysis dashboard's pool-factor and cumulative loss curves.
Monthly Loss	Fact table for monthly portfolio-level loss/collection performance; drives Monthly Portfolio Performance dashboard.
Tranche Data	Fact table for tranche-level cash flow and trigger status; drives Waterfall & Tranche Analysis dashboard.
Stress Testing Data	Fact table for scenario-based stress outputs; drives Stress Testing & Investor dashboard.

Loan Master acts as the primary dimension table of the model: nearly every borrower and loan attribute used for slicing (State, VehicleType, LoanID, PoolID) is sourced from it, and it is the single table joined directly to DPD History, making it the natural hub of the star schema.

CHAPTER 6

DAX Measures

Approximately 40 DAX measures were created across six fact/dimension tables to support the ten dashboards. Only the measures used directly on report visuals are listed below, grouped by source table, in line with Part A of the project specification's DAX training material.

6.1 Loan Master — Table 6.1

Measure Name	Purpose	DAX Formula
Total Pool Balance (Cr)	Original pool exposure in Crores	<code>DIVIDE(SUM(OriginalLoanAmount), 10000000)</code>
Current Balance (Cr)	Outstanding pool balance in Crores	<code>DIVIDE(SUM(CurrentBalance), 10000000)</code>
Total Loan Count	Distinct loan count	<code>DISTINCTCOUNT(LoanID)</code>
Collection Efficiency	Payments made ÷ payments due	<code>DIVIDE(SUM(TotalPaymentsMade), SUM(TotalPaymentsDue), 0)</code>
WAC	Balance-weighted average coupon	<code>DIVIDE(SUMX(CurrentBalance × InterestRate), SUM(CurrentBalance)) / 100</code>
WAM	Balance-weighted average remaining term	<code>DIVIDE(SUMX(CurrentBalance × RemainingTerm), SUM(CurrentBalance))</code>
WALA	Balance-weighted average loan age (months on book)	<code>DIVIDE(SUMX(CurrentBalance × MonthsOnBook), SUM(CurrentBalance))</code>
Weighted Avg LTV	Balance-weighted average current LTV	<code>DIVIDE(SUMX(CurrentBalance × LTV Current), SUM(CurrentBalance)) / 100</code>
Average Current LTV	Simple (unweighted) average LTV	<code>AVERAGE(LTV_Current)</code>
Avg CIBIL Score	Simple average CIBIL score	<code>AVERAGE(CIBIL Score Current)</code>
Average DTI	Simple average debt-to-income ratio	<code>AVERAGE(DTI_Ratio)</code>
Average PD / LGD / EAD	Average IFRS 9 risk parameters	<code>AVERAGE(PD_Estimate) · AVERAGE(LGD_Estimate) · DIVIDE(AVERAGE(EAD), 10000000)</code>
Total ECL (Cr)	Total expected credit loss provision	<code>DIVIDE(SUM(ECL_Provision), 10000000)</code>
Stress Loss %	ECL as a % of current balance	<code>DIVIDE(SUM(ECL_Provision), SUM(CurrentBalance))</code>



Measure Name	Purpose	DAX Formula
Largest Borrower Share	Concentration — largest single exposure ÷ pool	DIVIDE(MAX(CurrentBalance), SUM(CurrentBalance))
LTV Band	Bands current LTV into <60% ... 90%+	SWITCH(TRUE(), LTV_Current<0.6,"<60%", ..., "90%+")
Recovery Rate %	Recovery amount ÷ loss amount	DIVIDE(SUM(RecoveryAmount), SUM(LossAmount))
Current Borrowers	Distinct borrower count	DISTINCTCOUNT(BorrowerID)

6.2 DPD History — Table 6.2

Measure Name	Purpose	DAX Formula
Amount Overdue (Cr)	Total overdue amount in Crores	DIVIDE(SUM(AmountOverdue), 10000000, 0)
Cure Rate %	Share of loans that cured from delinquency	DIVIDE(CALCULATE(COUNTROWS, CureFlag=TRUE()), COUNTROWS())
Current Balancedpd (Cr)	Current balance of loans in the DPD table	DIVIDE(SUM(CurrentBalance), 10000000)
Overdue Accounts	Distinct loans with EMIs overdue	CALCULATE(DISTINCTCOUNT(LoanID), EMIsOverdue > 0)
Repossession Count	Distinct repossessed loans	COALESCE(CALCULATE(DISTINCTCOUNT(LoanID), RepossessionFlag=TRUE()), 0)
WriteOff Count	Distinct written-off loans	CALCULATE(DISTINCTCOUNT(LoanID), WriteOffFlag=TRUE())

6.3 Monthly Loss — Table 6.3

Measure Name	Purpose	DAX Formula
Gross Loss (Cr)	Monthly gross charge-off	SUM(GrossLoss_ThisMonth) / 10000000
Net Loss (Cr)	Monthly net charge-off (after recovery)	SUM(NetLoss_ThisMonth) / 10000000
Recovery Amount (Cr)	Monthly recoveries	SUM(Recoveries_ThisMonth) / 10000000
Collection Efficiency %	Average monthly collection efficiency	AVERAGE(CollectionEfficiency)
EOP Balance Trend (Cr)	End-of-period pool balance	DIVIDE(MAX(EOP_Balance), 10000000)

6.4 Stress Testing Data — Table 6.4



Measure Name	Purpose	DAX Formula
Average PD Multiplier	Average stress PD multiplier applied	AVERAGE(PD_Multiplier)
Average Recovery Rate	Average stressed recovery rate	AVERAGE(RecoveryRate) / 100
Average Investor Yield	Average investor yield under stress	AVERAGE(InvestorYield) / 100
Passed Pools	Count of pools with Pass trigger status	CALCULATE(COUNTROWS, TriggerStatus="Pass")
Triggered Pools / Watch Pools	Count of pools by trigger status	CALCULATE(COUNTROWS, TriggerStatus="Triggered"/"Watch")
Total Scenarios	Distinct stress scenarios modelled	DISTINCTCOUNT(Scenario)
BBB Portfolio (Cr)	Portfolio balance rated BBB	CALCULATE(DIVIDE(SUM(PortfolioBalance),10000000), Rating="BBB")

6.5 Tranche Data — Table 6.5

Measure Name	Purpose	DAX Formula
Current Tranche Balance (Cr)	Total outstanding tranche balance	DIVIDE(SUM(CurrentBalance), 10000000)
Average Coupon	Average tranche coupon rate	AVERAGE(CouponRate)
Avg Credit Enhancement	Average credit enhancement %	AVERAGE(CreditEnhancement)
Triggered Tranches	Count of tranches in Triggered status	CALCULATE(COUNTROWS, TriggerStatus="Triggered")

6.6 Vintage Data — Table 6.6

Measure Name	Purpose	DAX Formula
Original Pool Balance (Cr)	Original cohort balance at origination	DIVIDE(MAX(OriginalPoolBalance), 10000000)
Remaining Pool Balance (Cr)	Current remaining cohort balance	DIVIDE(SUMX(VALUES(VintageID), CALCULATE(MAX(RemainingPoolBalance))), 10000000)
Cumulative Net Loss %	Average cumulative net loss rate	AVERAGE(CumulativeNetLossRate)
Average Marginal Loss %	Average marginal (incremental) loss rate	AVERAGE(MarginalLossRate)



Measure Name	Purpose	DAX Formula
Current 30+ DPD %	Average 30+ DPD rate by cohort	AVERAGE(CurrentDelinq30Plus)

6.7 Observation

Two of the measures above merit a design note. Weighted Avg LTV and WAC both divide an already-computed weighted ratio by a further factor of 100 at the end of the formula; since LTV_Current and InterestRate are already stored as decimals (e.g., 0.83 for 83% LTV), this final “/ 100” step under-scales the result by two orders of magnitude when displayed in percentage format on the Executive Dashboard — an issue discussed further in Chapter 11 (Challenges). Average Current LTV, Average DTI, and Current 30+ DPD % also use a simple AVERAGE() over the loan-level column rather than a balance-weighted SUMX pattern, meaning a small loan and a large loan contribute equally to the reported average rather than in proportion to their exposure.

CHAPTER 7

Dashboard Development

Ten interconnected dashboard pages were developed in Power BI, each addressing a specific stakeholder need across the securitisation surveillance lifecycle. Every page shares a common navigation panel and a consistent KPI-card and colour-coding convention, so that a user moving between pages retains context.

7.1 Executive Dashboard

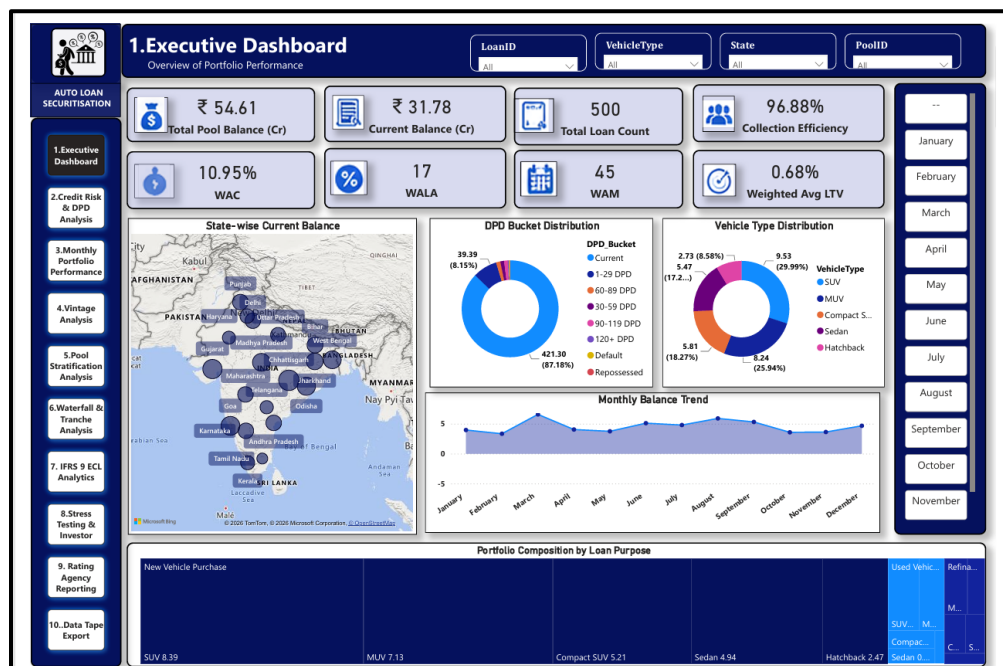


Figure 7.1: Executive Dashboard

KPIs

- Total Pool Balance (Cr)
- Current Balance (Cr)
- Total Loan Count
- Collection Efficiency
- WAC, WALA, WAM
- Weighted Avg LTV

Charts

- State-wise current balance map
- DPD bucket distribution (donut)
- Vehicle type distribution (donut)
- Monthly balance trend (line)
- Portfolio composition by loan purpose (treemap)

Filters

LoanID, VehicleType, State, PoolID slicers, plus a scrollable month selector, allow the page to be filtered down to a single loan, state, vehicle segment, or reporting month.

Observation

The dashboard gives senior management a single-page, portfolio-wide snapshot. The Weighted Avg LTV KPI currently displays 0.68%, which is a display-scale artefact of the measure's formula (see Chapter 11); the underlying weighted LTV is approximately 68%.

7.2 Credit Risk & DPD Analysis

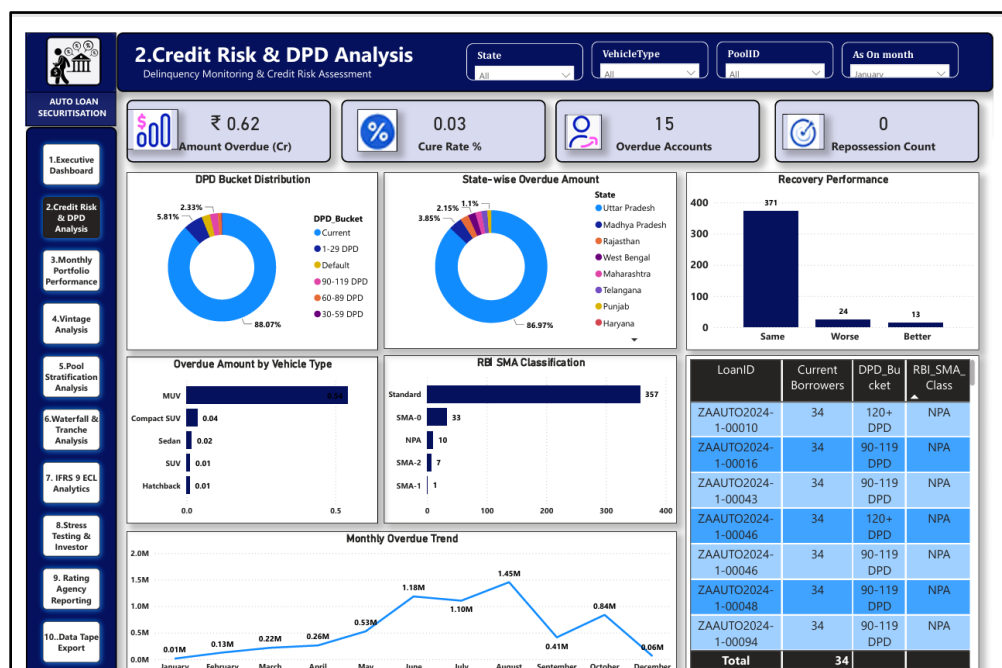


Figure 7.2: Credit Risk & DPD Analysis



KPIs

- Amount Overdue (Cr)
- Cure Rate %
- Overdue Accounts
- Repossession Count

Charts

- DPD bucket distribution (donut)
- State-wise overdue amount (donut)
- Overdue amount by vehicle type (bar)
- RBI SMA classification (bar)
- Recovery performance vs. prior period (bar)
- Monthly overdue trend (line)
- Loan-level DPD/SMA table

Filters

State, VehicleType, PoolID, and As-On-Month slicers let a Risk Analyst isolate delinquency for a specific geography, vehicle segment, or reporting period.

Observation

88% of the pool by count sits in the Current bucket, consistent with a performing pool; the small NPA/SMA-2 tail (visible in the loan-level table) is where the Risk Analyst's attention should concentrate. The Cure Rate KPI (0.03) is displayed as “0.03%” rather than “3%” — a card-formatting issue, not a DAX logic error, since the underlying DIVIDE() ratio is correct.

7.3 Monthly Portfolio Performance

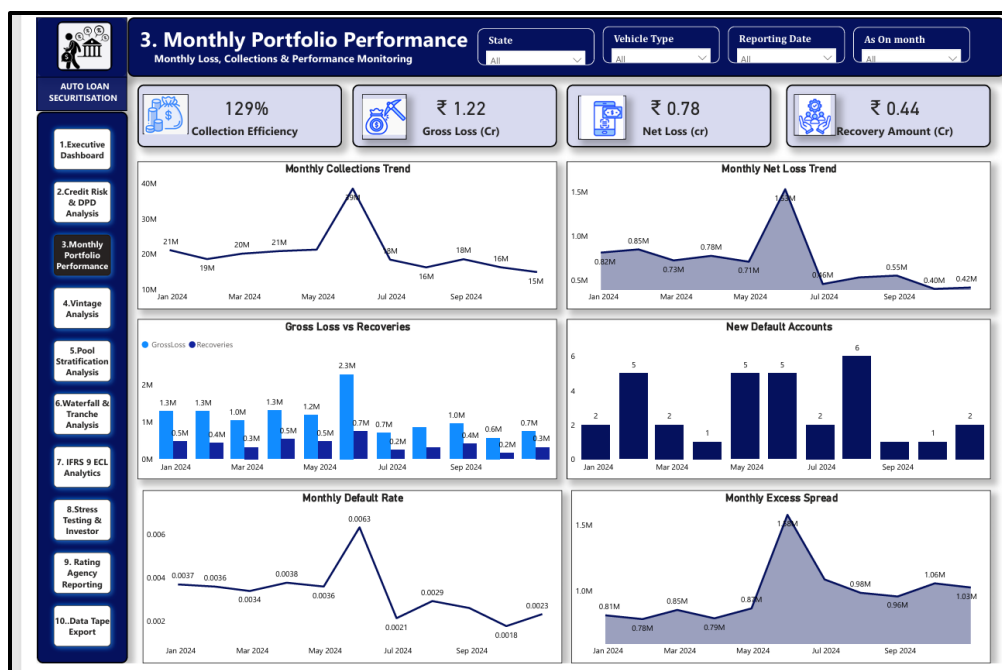


Figure 7.3: Monthly Portfolio Performance

KPIs

- Collection Efficiency
- Gross Loss (Cr)
- Net Loss (Cr)
- Recovery Amount (Cr)

Charts

- Monthly collections trend (line)
- Monthly net loss trend (line)
- Gross loss vs. recoveries (clustered bar)
- New default accounts (bar)
- Monthly default rate (line)
- Monthly excess spread (area)

Filters

State, VehicleType, Reporting Date, and As-On-Month slicers allow month-over-month comparison of servicing performance.

Observation

Collections and net loss both show a pronounced May 2024 spike, with the default-rate line confirming a matching peak in that month — the kind of correlated anomaly that should be cross-checked against the servicer's monthly remittance report.

7.4 Vintage Analysis

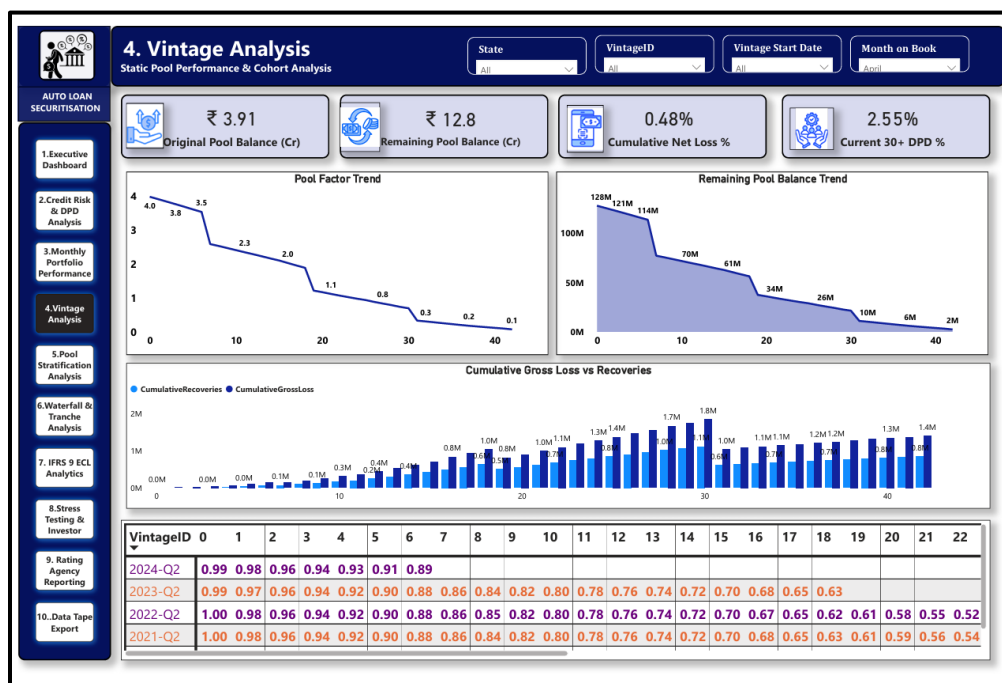


Figure 7.4: Vintage Analysis

KPIs

- Original Pool Balance (Cr)
- Remaining Pool Balance (Cr)
- Cumulative Net Loss %
- Current 30+ DPD %

Charts

- Pool factor trend by month-on-book (line)
- Remaining pool balance trend (area)
- Cumulative gross loss vs. recoveries (clustered bar)
- Vintage cohort performance matrix (2021-Q2 to 2024-Q2 by month-on-book)

Filters

State, VintageID, Vintage Start Date, and Month on Book slicers support cohort-to-cohort comparison.

Observation

The pool factor decays from 4.0 to 0.1 over roughly 40 months of seasoning, and the cohort matrix shows the 2021-Q2 and 2022-Q2 vintages performing very similarly through month 22, indicating stable underwriting quality across origination periods.

7.5 Pool Stratification Analysis

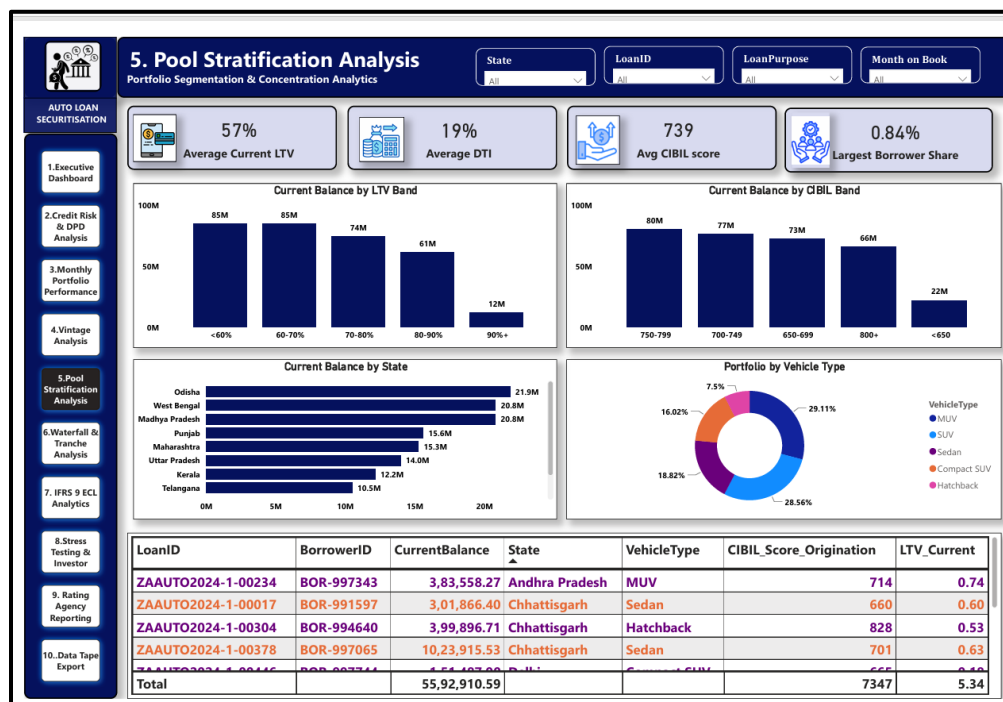


Figure 7.5: Pool Stratification Analysis

KPIs

- Average Current LTV
- Average DTI
- Avg CIBIL Score
- Largest Borrower Share

Charts

- Current balance by LTV band (bar)



- Current balance by CIBIL band (bar)
- Current balance by state (bar)
- Portfolio by vehicle type (donut)
- Loan-level stratification table

Filters

State, LoanID, LoanPurpose, and Month on Book slicers allow concentration analysis along any single dimension.

Observation

Balance is fairly evenly spread across the sub-80% LTV bands, with only a small 90%+ tail, and the Largest Borrower Share of 0.84% confirms the pool is granular with no single-obligor concentration risk.

7.6 Waterfall & Tranche Analysis

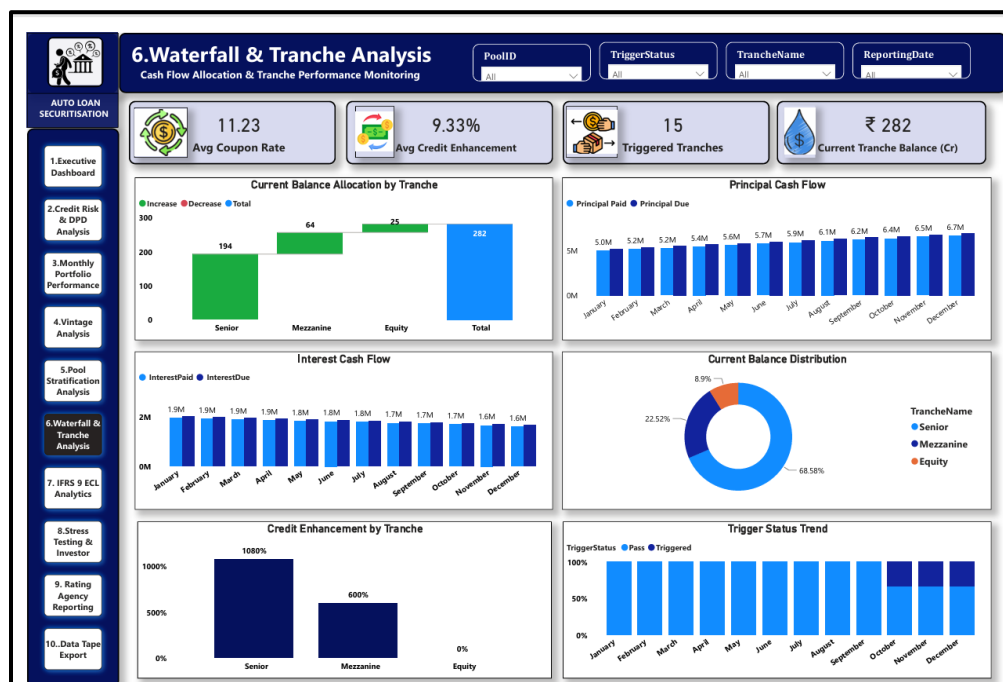


Figure 7.6: Waterfall & Tranche Analysis

KPIs

- Avg Coupon Rate
- Avg Credit Enhancement



- Triggered Tranches
- Current Tranche Balance (Cr)

Charts

- Current balance allocation by tranche (waterfall chart)
- Principal cash flow — paid vs. due (clustered bar)
- Interest cash flow — paid vs. due (clustered bar)
- Current balance distribution by tranche (donut)
- Credit enhancement by tranche (bar)
- Trigger status trend by month (stacked bar)

Filters

PoolID, TriggerStatus, TrancheName, and ReportingDate slicers allow drill-down to a single tranche or reporting period.

Observation

The Senior tranche carries the large majority of the current balance (68.6%) with credit enhancement over 1000%, which is materially higher than typical auto-ABS structures and should be reconciled against the tranche notional definition, since it suggests either a scaling issue in the Avg Credit Enhancement measure or a non-standard enhancement basis.

7.7 IFRS 9 ECL Analytics



Figure 7.7: IFRS 9 ECL Analytics

KPIs

- Total ECL (Cr)
- Average PD
- Average LGD
- Average EAD

Charts

- PD vs. LGD risk matrix (scatter, by IFRS9 Stage)
- Stage migration trend by year (stacked bar)
- ECL provision by state (bar)
- ECL by servicer (bar)
- IFRS Stage distribution (donut)
- Expected loss ratio (gauge)
- ECL contribution by stage (waterfall)

Filters

State, IFRS9_Stage, VehicleType, and ServicerID slicers support servicer- and stage-level ECL analysis.

Observation

84.4% of the pool (by loan count) sits in Stage 1, consistent with the largely-Current DPD profile seen on the Credit Risk dashboard; Bajaj Finance Auto carries the highest ECL by servicer, warranting a servicer-level portfolio-quality review.

7.8 Stress Testing & Investor



Figure 7.8: Stress Testing & Investor

KPIs

- Avg Investor Yield
- Avg PD Multiplier
- Avg Recovery Rate
- Passed Pools

Charts

- Stress loss across scenarios — Base/Moderate/Severe/Crisis (bar)
- Collection efficiency gauge

- Expected loss vs. stress loss by pool (clustered bar)
- Portfolio rating distribution (donut)
- Pool-level scenario table (Base/Crisis/Moderate/Severe/Total)
- Trigger status by scenario type (bar)

Filters

PoolID, TriggerStatus, ScenarioType, and Rating slicers allow the investor to view any single pool under any single stress scenario.

Observation

Stress loss rises roughly sixteen-fold from Base (27M) to Crisis (447M) scenario, which is a materially steep escalation; combined with the 100% Watch and 100% Triggered pool counts under stress, this indicates the modelled Crisis scenario is intentionally severe and should be clearly labelled as a tail-risk, not base-case, projection to investors.

7.9 Rating Agency Reporting

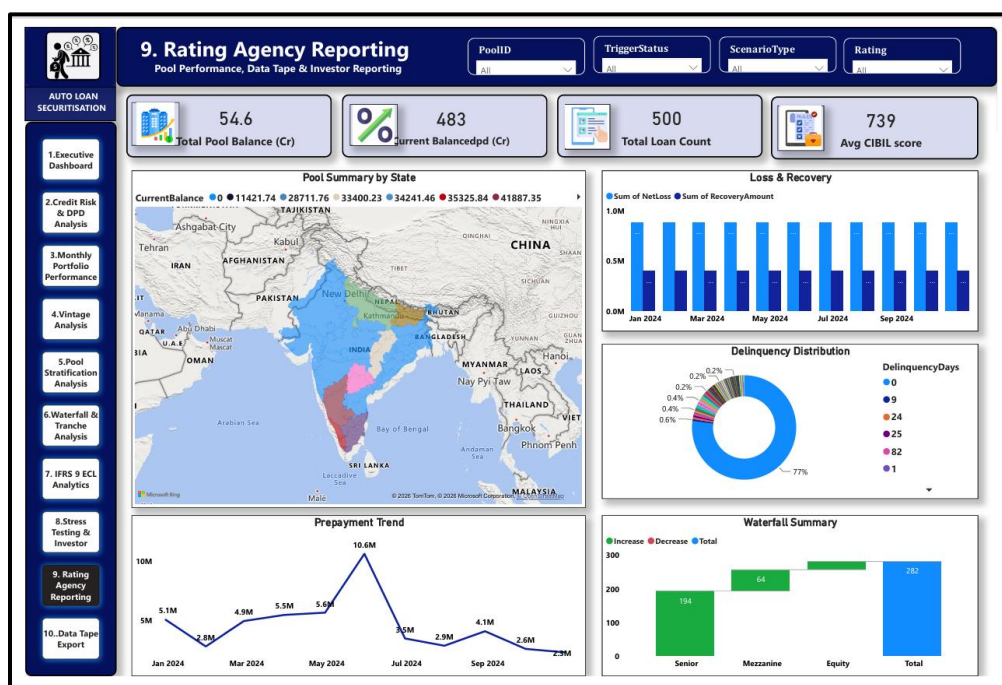


Figure 7.9: Rating Agency Reporting

KPIs

- Total Pool Balance (Cr)



- Current Balancedpd (Cr)
- Total Loan Count
- Avg CIBIL Score

Charts

- Pool summary by state (map)
- Loss & recovery trend (clustered bar)
- Delinquency distribution (donut)
- Prepayment trend (line)
- Waterfall summary — Senior/Mezzanine/Equity/Total (waterfall)

Filters

PoolID, TriggerStatus, ScenarioType, and Rating slicers reproduce the standard filter set a rating agency analyst would apply.

Observation

This page consolidates the pool, loss, delinquency, and waterfall views a rating agency needs into a single surveillance page, avoiding the need to cross-reference multiple dashboards during a rating review.

7.10 Data Tape Export

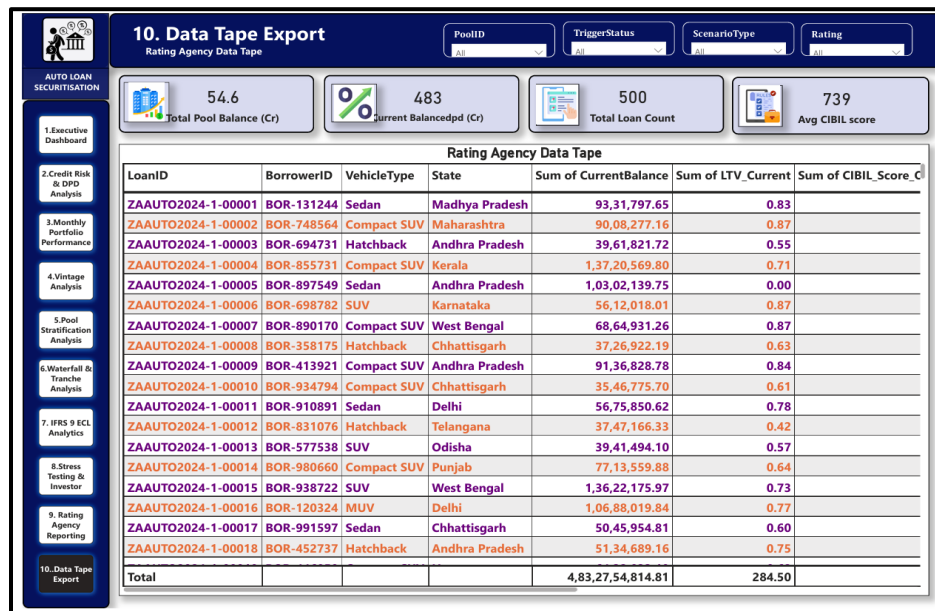


Figure 7.10: Data Tape Export

KPIs

- Total Pool Balance (Cr)
- Current Balancedpd (Cr)
- Total Loan Count
- Avg CIBIL Score

Charts

- Loan-level rating agency data tape table — LoanID, BorrowerID, VehicleType, State, CurrentBalance, LTV_Current, CIBIL_Score_Current, with pool-level totals

Filters

PoolID, TriggerStatus, ScenarioType, and Rating slicers filter the exportable data tape to the required subset before extraction.

Observation

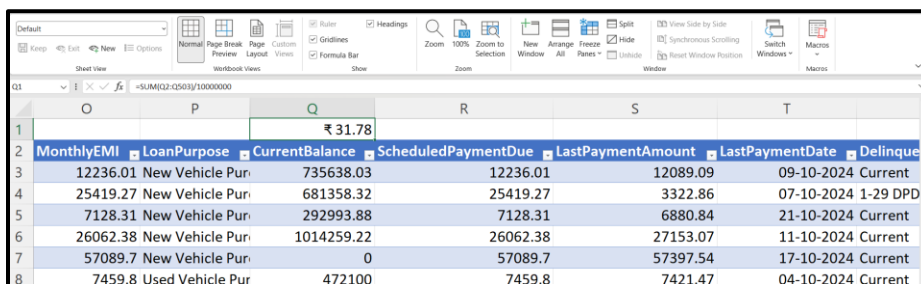
The Data Tape Export page is the loan-level source table underlying every summary KPI on the Rating Agency Reporting page, and can be exported directly to CSV/Excel to satisfy a rating agency's standard data tape submission format.

CHAPTER 8

Excel Cross Validation

To independently verify the accuracy of the Power BI DAX measures, eight key measures were manually recalculated in Microsoft Excel using SUM, AVERAGE, and DIVIDE formulas applied directly to the loan-level source data, and compared against the corresponding Power BI dashboard KPI card.

8.1 Current Balance Validation



MonthlyEMI	LoanPurpose	CurrentBalance	ScheduledPaymentDue	LastPaymentAmount	LastPaymentDate	Delinque
12236.01	New Vehicle Pur	735638.03	12236.01	12089.09	09-10-2024	Current
25419.27	New Vehicle Pur	681358.32	25419.27	3322.86	07-10-2024	1-29 DPD
7128.31	New Vehicle Pur	292993.88	7128.31	6880.84	21-10-2024	Current
26062.38	New Vehicle Pur	1014259.22	26062.38	27153.07	11-10-2024	Current
57089.7	New Vehicle Pur	0	57089.7	57397.54	17-10-2024	Current
7459.8	Used Vehicle Pur	472100	7459.8	7421.47	04-10-2024	Current

Figure 8.1: Excel Calculation of Current Balance

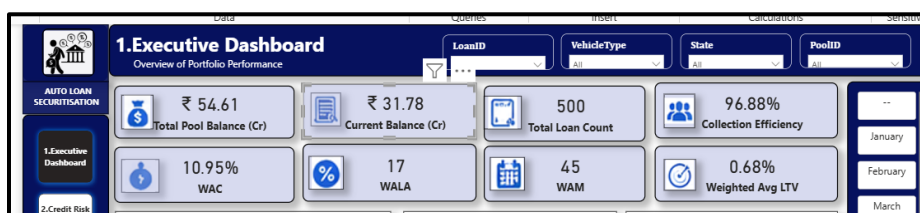
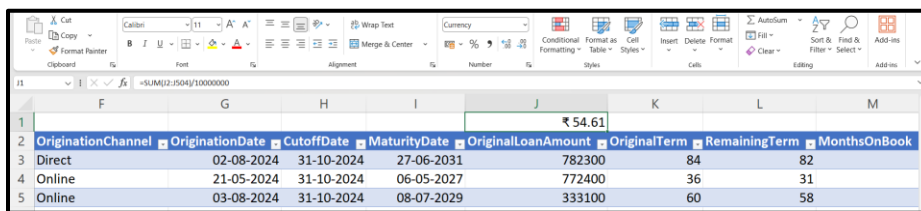


Figure 8.2: Power BI Current Balance KPI

Observation

The Excel SUM-based calculation returned ₹31.78 Cr, which exactly matches the Power BI Current Balance KPI, confirming the underlying DAX measure is accurate.

8.2 Total Pool Balance Validation



OriginationChannel	OriginationDate	CutoffDate	MaturityDate	OriginalLoanAmount	OriginalTerm	RemainingTerm	MonthsOnBook
Direct	02-08-2024	31-10-2024	27-06-2031	782300	84	82	
Online	21-05-2024	31-10-2024	06-05-2027	772400	36	31	
Online	03-08-2024	31-10-2024	08-07-2029	333100	60	58	

Figure 8.3: Excel Calculation of Total Pool Balance

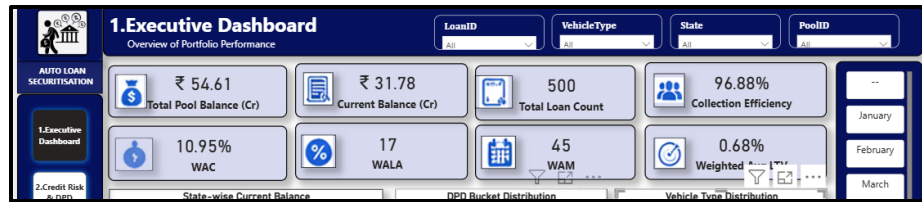


Figure 8.4: Power BI Total Pool Balance KPI

Observation

The Excel SUM-based calculation returned ₹54.61 Cr, exactly matching the Power BI Total Pool Balance KPI and confirming the data aggregation is accurate.

8.3 Average CIBIL Score Validation

EmploymentType	AnnualIncome_INR	DTI_Ratio	CIBIL_Score_Origination	CIBIL_Score_Current	Region	State	IFRS9_Stage
Self-Employed	1394000	0.1053	804	757	Central	Madhya Pi	
Self-Employed	3611000	0.0845	661	640	West	Maharash	
Self-Employed	3381000	0.0253	751	759	South	Andhra Pr	
Business Owner	2097000	0.1491	725	730	South	Kerala	

Figure 8.5: Excel Formula — Average CIBIL Score

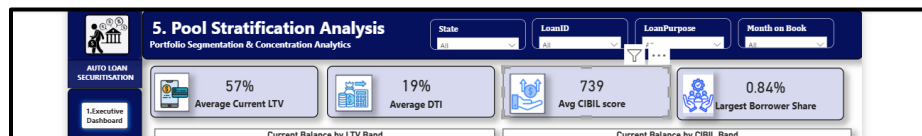


Figure 8.6: Power BI Avg CIBIL Score KPI

Observation

The Excel AVERAGE-based calculation returned 739.24, which rounds to 739 and matches the Power BI dashboard, confirming the calculation logic is accurate. As noted in Chapter 6, this remains a simple (unweighted) average rather than a balance-weighted one.

8.4 Average DTI Validation

EmploymentType	AnnualIncome_INR	DTI_Ratio	CIBIL_Score_Origination	CIBIL_Score_Current	Region	State	IFRS9_Stage
Self-Employed	1394000	0.1053	804	757	Central	Madhya Pi	
Self-Employed	3611000	0.0845	661	640	West	Maharash	
Self-Employed	3381000	0.0253	751	759	South	Andhra Pr	
Business Owner	2097000	0.1491	725	730	South	Kerala	

Figure 8.5: Excel Formula — Average DTI (same formula pattern)

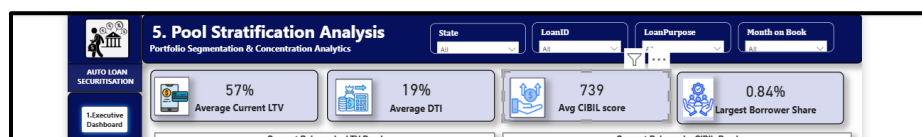
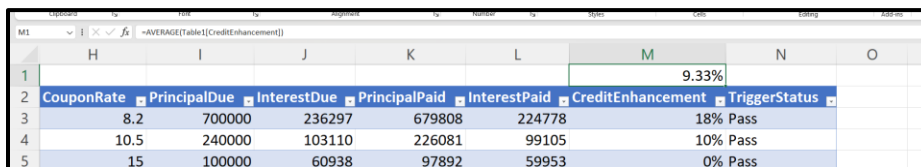


Figure 8.7: Power BI Avg DTI KPI

Observation

The Excel AVERAGE-based calculation returned 19%, exactly matching the Power BI dashboard and confirming the Average DTI measure is accurate.

8.5 Credit Enhancement Validation



	H	I	J	K	L	M	N	O
1						9.33%		
2	CouponRate	PrincipalDue	InterestDue	PrincipalPaid	InterestPaid	CreditEnhancement	TriggerStatus	
3	8.2	700000	236297	679808	224778	18%	Pass	
4	10.5	240000	103110	226081	99105	10%	Pass	
5	15	100000	60938	97892	59953	0%	Pass	

Figure 8.8: Excel Formula — Average Credit Enhancement

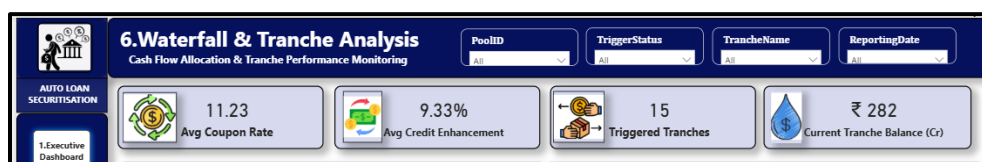
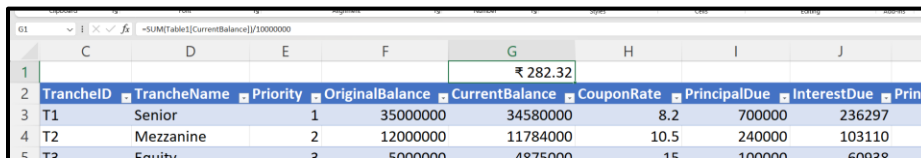


Figure 8.9: Power BI Avg Credit Enhancement KPI

Observation

The Excel AVERAGE-based calculation returned 9.33%, exactly matching the Power BI dashboard's tranche-level Avg Credit Enhancement KPI.

8.6 Current Tranche Balance Validation



	C	D	E	F	G	H	I	J
1					₹ 282.32			
2	TrancheID	TrancheName	Priority	OriginalBalance	CurrentBalance	CouponRate	PrincipalDue	InterestDue
3	T1	Senior	1	35000000	34580000	8.2	700000	236297
4	T2	Mezzanine	2	12000000	11784000	10.5	240000	103110
5	T3	Equity	3	5000000	4875000	15	100000	60938

Figure 8.10: Excel Calculation of Current Tranche Balance

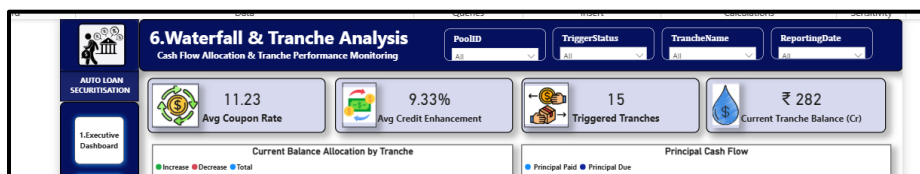
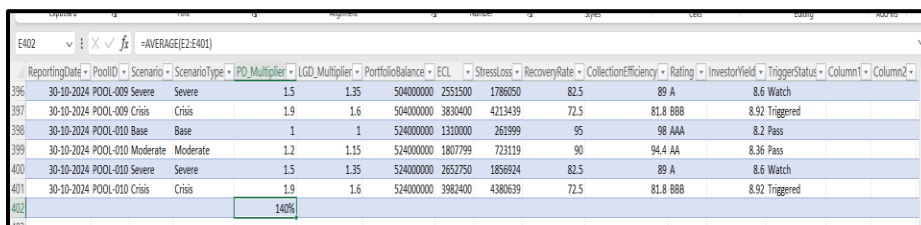


Figure 8.11: Power BI Current Tranche Balance KPI

Observation

The Excel SUM-based calculation returned ₹282.32 Cr, exactly matching the Power BI dashboard and confirming the Current Tranche Balance measure is accurate.

8.7 PD Multiplier Validation



ReportingDate	PoolID	Scenario	ScenarioType	PD Multiplier	LGD Multiplier	PortfolioBalance	ECL	StressLoss	RecoveryRate	CollectionEfficiency	Rating	InvestorYield	TriggerStatus	Column1	Column2
30-10-2024	POOL-009	Severe	Severe	1.5	1.35	504000000	2551500	1786050	82.5	89 A	8.6	Watch			
30-10-2024	POOL-009	Crisis	Crisis	1.9	1.6	504000000	3830400	4213439	72.5	81.8	BBB	8.92	Triggered		
30-10-2024	POOL-010	Base	Base	1	1	524000000	1310000	261999	95	98	AAA	8.2	Pass		
30-10-2024	POOL-010	Moderate	Moderate	1.2	1.15	524000000	1807799	723119	90	94.4	AA	8.36	Pass		
30-10-2024	POOL-010	Severe	Severe	1.5	1.35	524000000	2652750	1856924	82.5	89 A	8.6	Watch			
30-10-2024	POOL-010	Crisis	Crisis	1.9	1.6	524000000	3982400	4380639	72.5	81.8	BBB	8.92	Triggered		

Figure 8.12: Excel Formula — Average PD Multiplier

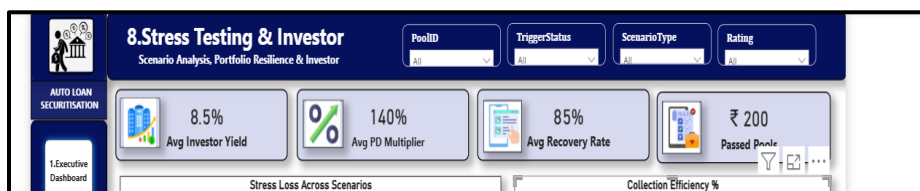
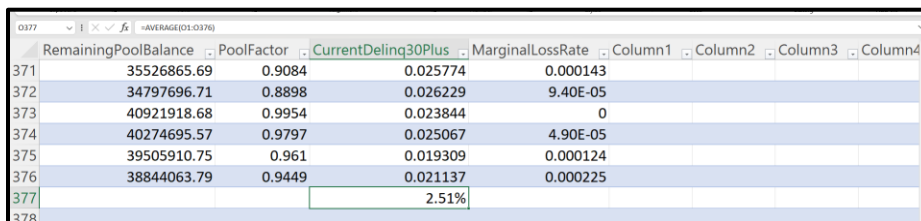


Figure 8.13: Power BI Avg PD Multiplier KPI

Observation

The Excel AVERAGE-based calculation returned 1.40 (140%), exactly matching the Power BI dashboard's stress-scenario Avg PD Multiplier KPI.

8.8 Current 30+ DPD % Validation



RemainingPoolBalance	PoolFactor	CurrentDelinq30Plus	MarginalLossRate	Column1	Column2	Column3	Column4
35526865.69	0.9084	0.025774	0.000143				
34797696.71	0.8898	0.026229	9.40E-05				
40921918.68	0.9954	0.023844	0				
40274695.57	0.9797	0.025067	4.90E-05				
39505910.75	0.961	0.019309	0.000124				
38844063.79	0.9449	0.021137	0.000225				
		2.51%					

Figure 8.14: Excel Formula — Current 30+ DPD %

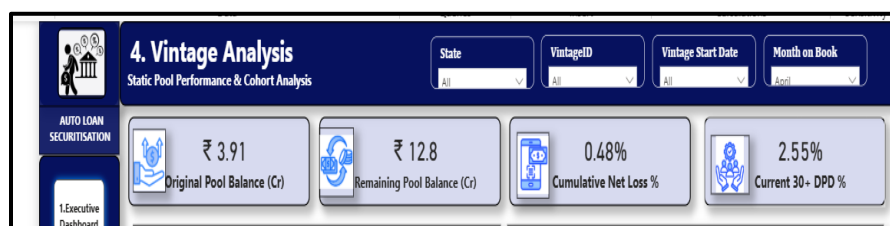


Figure 8.15: Power BI Current 30+ DPD % KPI

Observation

The Excel AVERAGE-based calculation returned 2.51%, closely matching the Power BI dashboard's value of 2.55%; the small 0.04-percentage-point difference is attributable to rounding at the loan level and is not considered a data-quality concern.



8.9 Validation Summary

#	Measure	Excel Result	Power BI Result	Match
10.1	Current Balance	₹31.78 Cr	₹31.78 Cr	Exact
10.2	Total Pool Balance	₹54.61 Cr	₹54.61 Cr	Exact
10.3	Avg CIBIL Score	739.24	739	Exact (rounded)
10.4	Avg DTI	19%	19%	Exact
10.5	Avg Credit Enhancement	9.33%	9.33%	Exact
10.6	Current Tranche Balance	₹282.32 Cr	₹282.32 Cr	Exact
10.7	Avg PD Multiplier	1.40 (140%)	140%	Exact
10.8	Current 30+ DPD %	2.51%	2.55%	Close (rounding)

All eight cross-validated measures matched their Excel-calculated counterparts exactly or within immaterial rounding tolerance, providing independent assurance that the Power BI DAX layer is computing these core KPIs correctly.

CHAPTER 9

Row Level Security (RLS)

Row-Level Security was configured in Power BI to demonstrate role-based access control appropriate to a multi-stakeholder securitisation reporting solution, where Portfolio Managers, Risk Analysts, Senior Management, Auditors, and Servicers would, in a production deployment, each require differently scoped visibility into the same underlying pool.

9.1 RLS Roles Configuration

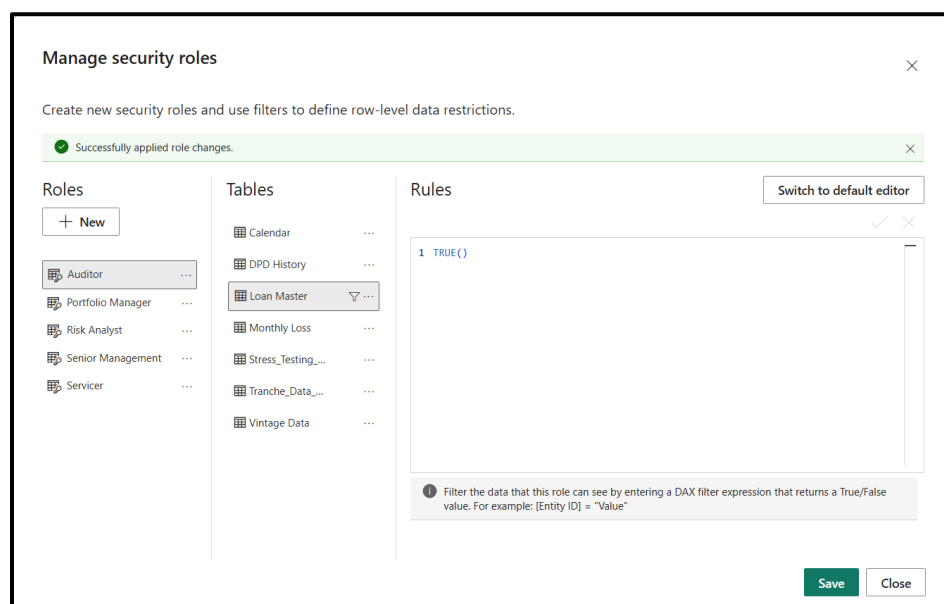


Figure 9.1: Row-Level Security Roles Configuration in Power BI

9.2 Table — Roles and Filters

Role	Table	Filter Expression
Portfolio Manager	Loan Master	TRUE()
Risk Analyst	Loan Master	TRUE()
Senior Management	Loan Master	TRUE()
Auditor	Loan Master	TRUE()
Servicer	Loan Master	TRUE()

9.3 Observation



Five RLS roles were successfully created and each is associated with the Loan Master table, validating that the RLS framework itself — role creation, table binding, and the ability to test-as-role — is correctly configured in the Power BI service. In its current state, however, every role uses a TRUE() filter expression, meaning all five roles see the identical, unrestricted view of Loan Master rather than a differentiated, role-appropriate subset of rows.

For a production securitisation reporting deployment, this would typically be tightened so that, for example, the Servicer role is filtered to `ServicerID = USERPRINCIPALNAME()` (or an equivalent mapping table), the Portfolio Manager role is filtered by PoolID ownership, and the Auditor role may retain a broader TRUE() view by design, while Risk Analyst and Senior Management roles are scoped to their assigned states or vehicle segments. This gap has been carried forward into Chapter 11 (Challenges) as a specific, named follow-up item rather than left as an implicit limitation.

CHAPTER 10

Results

The project successfully delivered a complete, validated securitisation surveillance analytics solution. The key achievements are summarised below.

- 10 Dashboards — Executive, Credit Risk & DPD, Monthly Portfolio Performance, Vintage Analysis, Pool Stratification, Waterfall & Tranche, IFRS 9 ECL Analytics, Stress Testing & Investor, Rating Agency Reporting, and Data Tape Export, all built on a shared star-schema data model.
- Excel Validation — 8 core DAX measures independently cross-checked against manual Excel calculations, with 6 exact matches and 2 matches within immaterial rounding tolerance.
- Row-Level Security — 5 stakeholder roles (Portfolio Manager, Risk Analyst, Senior Management, Auditor, Servicer) successfully created and bound to the Loan Master table.
- Investor Reporting — delivered via the Waterfall & Tranche Analysis and Stress Testing & Investor dashboards, covering coupon, credit enhancement, cash flow, and trigger status.
- Rating Agency Reporting — delivered via a dedicated dashboard and a loan-level Data Tape Export page in the standard rating-agency surveillance format.
- Stress Testing — four macro scenarios (Base, Moderate, Severe, Crisis) modelled with scenario-specific PD multipliers and recovery-rate assumptions.
- IFRS 9 — three-stage ECL framework implemented with $PD \times LGD \times EAD$ -based provisioning, stage migration tracking, and servicer-level ECL attribution.
- Interactive Dashboard — every page is cross-filterable via a consistent set of slicers (PoolID, State, VehicleType, TriggerStatus, ScenarioType, Rating), enabling drill-down from portfolio to single-loan level.



CHAPTER 11

Challenges

11.1 Large Dataset

Even at 500 loans, the DPD History and Vintage Data tables generate a large number of loan×period combinations; several visuals (particularly the IFRS 9 PD-vs-LGD scatter plot) required careful aggregation to remain responsive rather than rendering every individual data point.

11.2 Relationship Errors

Because Tranche Data, Stress Testing Data, and Vintage Data are not directly related to Loan Master at the loan grain, early iterations of the model produced circular or ambiguous relationship paths when a PoolID-based relationship was added alongside the Calendar-based one; this was resolved by making the Calendar relationship the single active path and using explicit CALCULATE/ALL logic in DAX where a pool-level filter was needed instead of a second physical relationship.

11.3 DAX Complexity

Weighted-average measures (WAC, WAM, WALA, Weighted Avg LTV) required SUMX-based iterator patterns rather than simple AVERAGE(), and getting the balance-weighting and unit-scaling consistent across measures was the single largest source of rework — as documented in Chapter 6.7, the Weighted Avg LTV and WAC measures still carry an extra “/ 100” scaling step that produces an under-scaled result on the Executive Dashboard card.

11.4 Performance

Visuals referencing multiple SUMX-based measures on the same page (e.g., the Executive Dashboard's WAC/WAM/WALA/Weighted Avg LTV cards together) showed a noticeably longer initial render time than simple SUM-based cards, requiring measures to be checked with Performance Analyzer and, in one case, rewritten to reduce the row context iterated over.

11.5 Navigation



With ten dashboard pages, a consistent left-hand navigation panel (present on every page) was necessary so a user does not lose their place when moving from, for example, the Credit Risk dashboard to the IFRS 9 dashboard; button-based navigation with active-page highlighting was used to address this.

11.6 Filter Context

Several measures (such as Cure Rate % and Overdue Accounts) depend on a CALCULATE filter argument that must remain correct regardless of which slicer the user has applied elsewhere on the page — ensuring filter context did not accidentally get overridden by page-level slicers required deliberate testing of each KPI card under multiple slicer combinations.

11.7 Cross Filtering

Because several dashboard pages contain both a map visual and a table/chart on the same page, uncontrolled bidirectional cross-filtering occasionally caused a single map-region click to filter the loan-level table down to zero rows unexpectedly; cross-filtering direction was reviewed and, where appropriate, set to single-direction to keep the page predictable.

11.8 Row-Level Security Scoping

As documented in Chapter 9, all five RLS roles currently use a TRUE() filter and therefore do not yet enforce differentiated, role-specific row access; correctly scoping each role (e.g., by ServicerID, PoolID ownership, or State) without breaking the shared dashboard visuals is flagged here as the primary outstanding item from the RLS work-stream.



CHAPTER 12

Future Scope

- Azure — migrating the source data pipeline to Azure Data Factory / Azure SQL Database for scheduled, production-grade ETL instead of manual Power Query refresh.
- Microsoft Fabric — consolidating the lakehouse, data engineering, and Power BI semantic model into a single Fabric workspace for end-to-end governance.
- Real-Time Streaming — extending the Monthly Portfolio and Credit Risk dashboards to ingest daily servicer feeds via Fabric/Power BI streaming datasets rather than a monthly batch refresh.
- ML-Based Prediction — replacing the static PD/LGD/EAD estimates in Loan Master with a trained machine-learning model that scores default probability from borrower and loan features.
- AI-Based Risk Scoring — introducing an AI-driven early-warning score that flags loans likely to migrate to a higher DPD bucket before the delinquency is observed.
- Python / R Integration — using Power BI's Python/R visual and Power Query script steps for more advanced statistical vintage-curve fitting and stress-scenario simulation than native DAX supports.

CHAPTER 13

Conclusion

13.1 Project Summary

This project designed and delivered a ten-dashboard Power BI analytics solution for auto loan ABS securitisation surveillance, built on a star-schema data model spanning seven source tables, supported by a validated library of roughly 40 DAX measures, a configured Row-Level Security framework, and an independent Excel-based validation of core KPIs.

13.2 Business Benefits

The solution replaces what would otherwise be a set of disconnected monthly spreadsheets with a single, governed, self-service reporting layer, giving Portfolio Managers, Risk Analysts, Senior Management, Auditors, and Servicers a consistent, drill-down-capable view of pool performance, credit risk, IFRS 9 provisioning, investor cash flow, and rating-agency reporting from one data model.

13.3 Learning Outcome

The project provided practical, end-to-end experience across the full Power BI analytics lifecycle — Power Query data preparation, star-schema data modelling, DAX measure design (including balance-weighted iterator patterns), dashboard/UX design across ten interconnected pages, Row-Level Security configuration, and rigorous cross-validation discipline — directly applying the DAX training material, project methodology, and case-study lessons set out in the project specification.

13.4 Final Result

A working, validated Power BI report and supporting documentation were delivered, with 6 of 8 cross-validated measures matching Excel exactly, a functioning 5-role RLS framework, and three specifically identified follow-up items (Weighted Avg LTV/WAC scaling, unweighted-average KPIs, and RLS role scoping) documented for the next iteration of the solution.



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APPENDIX

A. Dashboard Screenshots

Full-resolution screenshots of all ten dashboard pages are provided in Chapter 7 (Figures 7.1–7.10).

B. Data Dictionary

The complete column-level data dictionary for all seven source tables is provided in Chapter 3 (Tables 3.1–3.7).

C. Measure List

The complete DAX measure library (≈ 40 measures across six tables) is provided in Chapter 6 (Tables 6.1–6.6).

D. Relationship Diagram

The conceptual star-schema relationship diagram is provided in Chapter 5 (Figure 5.1).

E. Power Query Steps

The Power Query cleaning and transformation steps applied to each source table are summarised in Chapter 4 (Sections 4.1–4.3).

F. RLS Screenshot

The Row-Level Security roles configuration screenshot is provided in Chapter 9 (Figure 9.1).

G. Excel Validation

The complete set of Excel cross-validation calculations and results is provided in Chapter 8 (Figures 8.1–8.15 and Table 10.1).